FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – IV
PAPER –1
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –03, 18 – 20, 35 – 37):* This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(04 – 07, 21 – 24, 38 – 41)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(08 – 11, 25 – 28, 42 – 45):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(12 – 17, 29 – 34, 46 – 51):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

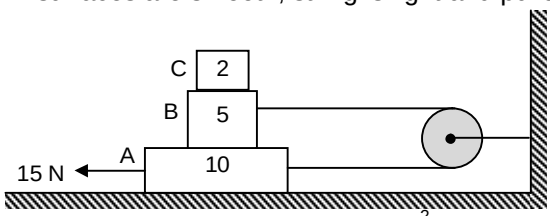
Physics

PART – I

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. All surfaces are smooth, string is light and pulley is frictionless, Mark the correct statement(s):



(A) Acceleration of block A is 1 m/s^2
(B) Acceleration of block A and B will be same
(C) Acceleration of block C is zero
(D) Tension in string between pulley and wall is 10 N.

2. Steel blocks A and B, which have equal masses, are at $T_A = 300^oC$ and $T_B = 400^oC$. Block C also of steel, with $m_C = 2m_A$ is at $T_C = 350^oC$. Blocks A and B are placed in contact, allowed to come to equilibrium and isolated. Then they are placed in contact with block C. After equilibrium if final temperature of A, B and C are T_A', T_B' and T_C' respectively:

(A) $T'_A = \sqrt{T'_B T'_C}$ (B) $T'_A = \dfrac{T'_B T'_C}{T'_B + T'_C}$

(C) $\dfrac{T'_A + T'_B}{2} = T'_C$ (D) $2T'_A - T'_B = T'_C$

3. An electric dipole moment $\vec{p} = \left(2.0\hat{i} + 3.0\hat{j}\right)\mu C$-m is placed in a uniform electric field $\vec{E} = \left(3.0\hat{i} + 2.0\hat{j}\right) \times 10^5 NC^{-1}$.

(A) The torque that $\vec{E}$ exerts on $\vec{P}$ is $\left(0.6\hat{i} - 0.4\hat{j} - 0.9\hat{k}\right)$Nm.

(B) The torque that $\vec{E}$ exerts on $\vec{p}$ is $\left(0.6\hat{i} - 0.4\hat{j} - 0.9\hat{k}\right)$Nm.

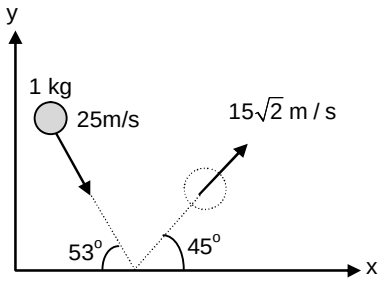
(C) The potential energy of the dipole is 0.6 J

(D) If the dipole is rotated in the electric field, the maximum potential energy of the dipole is 1.3 J.

SECTION – A
(One Options Correct Type)

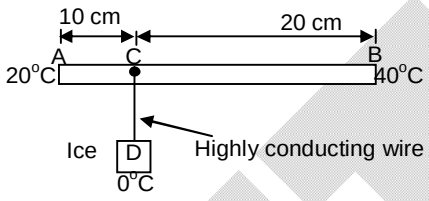
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. A ball of mass 1 kg bounces against the ground as shown in the figure. The approaching velocity is 25 m/s and the velocity after hitting the ground is $15\sqrt{2}$ m/s, the impulse exerted on the ball is:



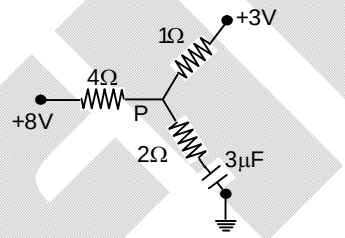
(A) 5 Ns
(B) $5\left(5-3\sqrt{2}\right)$Ns
(C) 35 Ns
(D) None of these

5. In the figure shown AB is a rod of length 30 cm and area of cross-section 1.0 cm^2 and thermal conductivity 336 S.I. units. The ends A and B are maintained at temperatures 20°C and 40°C respectively. A point C of this rod is connected to a box D, containing ice at 0°C, through a highly conducting wire of negligible heat capacity. The rate at which ice melts in the box is: [Assume latent heat of fusion for ice $L_F = 80$ cal / gm]



(A) 84 mg/s
(B) 84 g/s
(C) 20 mg/s
(D) 40 mg/s

6. The energy stored in the capacitor in the steady state is:



(A) 6μJ
(B) 24μJ
(C) 96μJ
(D) None of these

7. A plane loop of wire is placed in a region where the magnetic field is perpendicular to the plane of the loop and has the same magnitude and direction at all points within the area of the loop at any time. The magnitude of the magnetic field B varies with time according to the expression

$B = B_0\, e^{-at}$

where B_0 is maximum value of magnetic field and a is a positive constant. The time at which induced emf in the loop is maximum when:

(A) t = 0 (B) $t = \frac{1}{a}$

(C) $t = \frac{2}{a}$ (D) $t = \frac{1}{a}\ln 2$

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

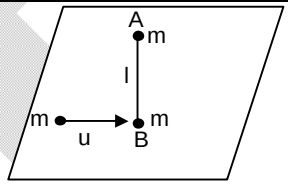
8. A particle of mass 1 kg is moving along a circle of radius 1 m. Then match the List - I with List – II.

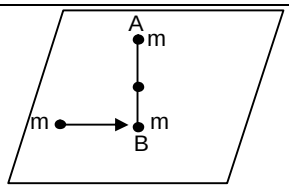
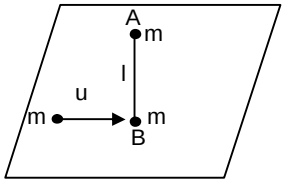
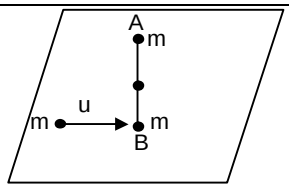
List - I		List - II	
(P)	Speed (v) of particle and time (t) are related as $v = 2t^2$.	(1)	Work done by net force on particle is positive.
(Q)	Kinetic energy and distance covered (s) by particle are related as $KE = 2s^2$.	(2)	Power supplied by net force is zero.
(R)	Angular velocity (ω) and time (t) are related as $\omega = 2t$.	(3)	Angle between net acceleration and line joining particle and center of circle changes continuously.
(S)	Angular displacement (θ) and time (t) are related as $\theta = 2 + 3t$	(4)	Magnitude of net acceleration changes continuously.
		(5)	Power supplied by net force is non-zero.

Codes:

	P	Q	R	S
(A)	1,2,3	2,3,5	1,3,4,5	5
(B)	1,3,4,5	1,3,4,5	1,3,4,5	2
(C)	1,2,3,4	1,3,5	2,3,4	1,3,4,5
(D)	1,3,4	2,3,4	1,3,5	2,3,4

9. Two small balls A and B, each of mass m, are joined rigidly to the ends of a light rigid rod of length l. The system lies on a smooth horizontal table. A particle of same mass moving with speed u perpendicularly to rod on the table strikes particle B, as shown, consider the following four situations and find the angular velocity ω of rod after collision.

List - I		List - II	
(P)	 Rod AB lies freely, collision is elastic	(1)	$\omega = \frac{u}{l}$

(Q)	 Rod AB is hinged in the middle, collision is elastic	(2)	$\omega = \frac{u}{2l}$
(R)	 Rod AB lies freely, colliding particle sticks	(3)	$\omega = \frac{4u}{3l}$
(S)	 Rod AB is hinged in the middle, colliding particle sticks	(4)	$\omega = \frac{2u}{3l}$
		(5)	$\omega = \frac{2u}{l}$

Codes:

	P	Q	R	S
(A)	1	3	2	4
(B)	4	5	2	3
(C)	2	1	5	3
(D)	5	3	2	1

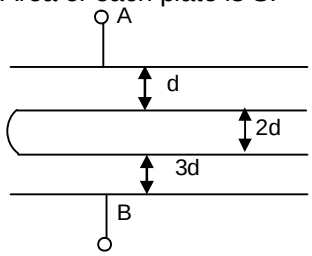
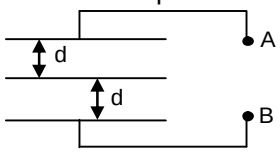
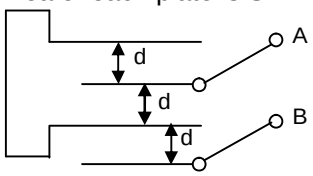
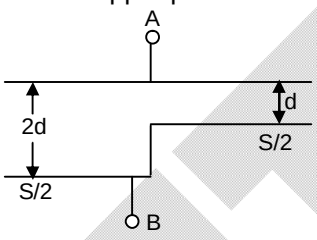
10. A spherical metallic conductor has a spherical cavity. A positive charge is placed inside the cavity at its center. Another positive charge is placed outside it. The conductor is initially electrically neutral.

List - I		List - II	
(P)	If outside charge is shifted to other position	(1)	Distribution of charge on inner surface of cavity changes
(Q)	If inside charge is shifted to other position within cavity	(2)	Distribution of charge on outer surface of conductor changes
(R)	If magnitude of charge inside cavity is increased	(3)	Electric potential at center of conductor due to charge present on outer surface of conductor changes
(S)	If conductor is earthed	(4)	Force on the charge placed inside cavity changes
		(5)	None

Codes:

	P	Q	R	S
(A)	1,2,3	4	1,4	5
(B)	2	1,4	1,2,3	2,3
(C)	2,3	1	1,3	1,2
(D)	1	5	1,4	3

11. The equivalent capacitance across AB for the plates connected as shown in column-I, are given in column-II. Match the items in column-II with the items in column-I.

List - I		List - II	
(P)	Area of each plate is S. 	(1)	$\frac{3\varepsilon_0 S}{4d}$
(Q)	Area of each plate is S. 	(2)	$\frac{2\varepsilon_0 S}{3d}$
(R)	Area of each plate is S. 	(3)	$\frac{\varepsilon_0 S}{2d}$
(S)	Area of upper plate is S. 	(4)	$\frac{\varepsilon_0 S}{2d}$
		(5)	$\frac{\varepsilon_0 S}{d}$

Codes:

	P	Q	R	S
(A)	4	5	2	3
(B)	3	2	4	1
(C)	2	3	1	5
(D)	3	4	2	1

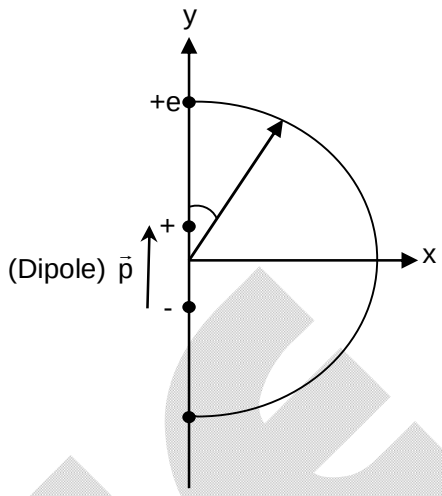
SECTION – B

(Numerical Answer Type)

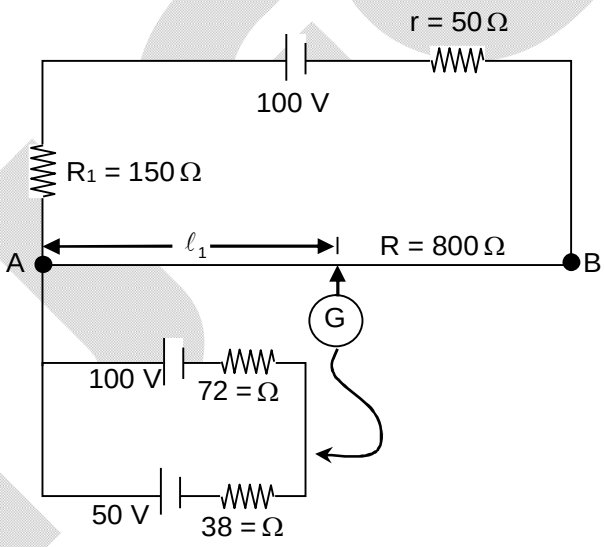
*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

12. A block of mass $m = \frac{1}{3}$ kg is kept on a rough horizontal plane. Friction coefficient is $\mu = 0.75$. The work done by minimum force required to drag the block along the plane by a distance 5 m, is W joule, then find the value of W.

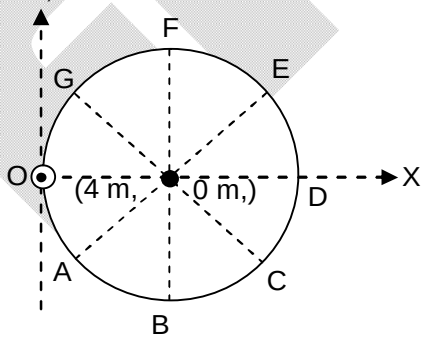
13. In the figure shown below, a particle of charge +e is initially at coordinate 16 nm on the dipole axis. The origin of y is at the dipole center. The particle is then moved along a circular path around the dipole center until it is at coordinate y = - 16 nm. The work done by the external force in moving the particle slowly is $-9\times10^{-30}\,J$. Find dipole moment and express your answer in multiple of 10^{-38} C-m.
$\left(e=1.6\times10^{-19}\,C\right)$



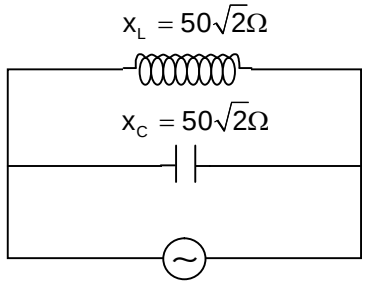
14. In the circuit shown length of A is 100 cm. Then find the value of ℓ_1 (in cm), where ℓ_1 corresponds null point.



15. An infinite uniform current carrying wire is kept along z-axis, carrying current I_0 in the direction of the positive z-axis. OABCDEFG represents a circle (where all the points are equally spaced), whose center at point (4 m, 0 m) and radius 4 m as shown in the figure. If $\int\limits_{DEF}\vec{B}\cdot\vec{d}\,l=\frac{\mu_0 I_0}{k}$ in S.I. unit, then find the value of k.



16. In an AC circuit capacitive reactance of capacitor is $X_C = 50\sqrt{2}\Omega$ and inductive reactance of inductor is $X_L = 50\sqrt{2}\Omega$. If the source voltage is 100 volts. Then find the current passing through the source.



17. A point source of sound is located somewhere along the x-axis. Experiments show that the same wavefront simultaneously reaches listeners at $x = -8$ m and $x = +2.0$ m. A third listener is positioned along the positive y-axis. What is his y co-ordinate (in m) if the same wavefront reaches him at the same instant as it does the first two listeners?

Chemistry

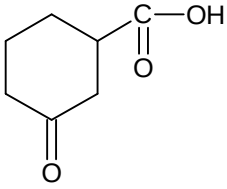
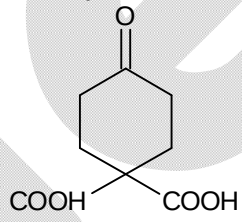
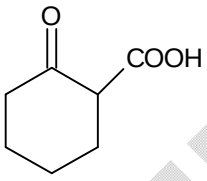
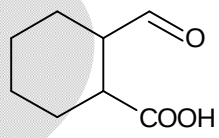
PART – II

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. Which of the following pairs form the same osazone with phenyl hydrazine?
(A) D - glucose & D – fructose
(B) D – fructose & D – mannose
(C) D – Glucose & D – Mannose
(D) D – glucose & D – galactose

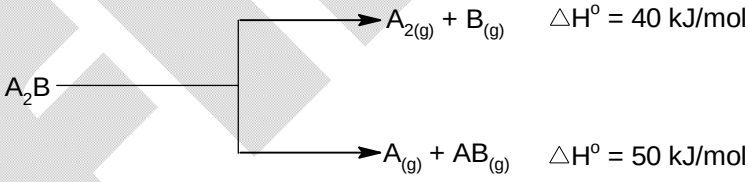
19. By which of the following reaction can MIC. (methyl Isocyanate) be obtained?
(A) $CH_3 - \overset{\oplus}{N} \equiv \overset{\ominus}{C} + HgO \rightarrow$
(B) $CH_3 - \overset{\oplus}{N} \equiv \overset{\ominus}{C} + O_3 \rightarrow$
(C) $CH_3 - \overset{\oplus}{N} \equiv \overset{\ominus}{C} + S \rightarrow$
(D) $CH_3 - \overset{\oplus}{N} \equiv \overset{\ominus}{C} + DMSO \rightarrow$

20. Which of the following compound easily undergoes decarboxylation on heating?
(A) 
(B) 
(C) 
(D) 

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. Substance A_2B can undergo decomposition to form two set of product.



If the molar ratio of $A_2(g)$ to $A(g)$ is 5 : 3 in a set of product gases, then energy involved in the decomposition of 1 mole of $A_2B(g)$ is
(A) 48.75 kJ/mol
(B) 43.73 kJ/mol
(C) 46.25 kJ/mol
(D) None of these

22. The volume of a colloidal particle is Vc. Volume of a solute particle in a true solution is Vt, the volume of suspension particle is Vs can be arranged.
(A) $Vc = Vt = Vs$
(B) $Vs < Vc < Vt$
(C) $Vs > Vc > Vt$
(D) $Vc > Vs > Vt$

23. Balmer gave an equation for wavelength of visible region of H-spectrum as $\lambda = \frac{Kn^2}{n^2-4}$. Where n = principal quantum numbers of energy level, K = constant in term of R, the value of K in term of R is

(A) R (B) R/2

(C) $\frac{4}{R}$ (D) $\frac{5}{R}$

24. A balloon of diameter 21 meter weights 100 kg calculate its pay load, if it is filled with He (gas) at 1.0 atm & 27°C. Density of air is 1.2 kg/m^3 given [R = 0.0821 lit atm k^{-1} mol^{-1}]

(A) 4952.42 kg (B) 4932.42 kg

(C) 493.242 kg (D) 2.24 kg

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match List – I with List – II:

List - I (Reaction) [X] = 1 M initially		List - II (If α is negligiable w.r.t. 1)	
(P)	$2X(g) \rightleftharpoons Y(g) + Z(g)$	(1)	$\alpha = 2 \times \sqrt{K_c}$
(Q)	$X(g) \rightleftharpoons Y(g) + Z(g)$	(2)	$\alpha = 3 \times \sqrt{K_c}$
(R)	$3X(g) \rightleftharpoons Y(g) + Z(g)$	(3)	$\alpha = (2K_c)^{1/3}$
(S)	$2X(g) \rightleftharpoons Y(g) + 2Z(g)$	(4)	$\alpha = \sqrt{K_c}$
		(5)	$\alpha = (K_C)^{1/3}$

Codes:

	P	Q	R	S
(A)	1	2	4	5
(B)	2	1	5	4
(C)	1	4	3	5
(D)	1	4	2	3

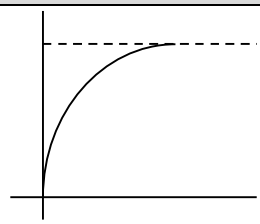
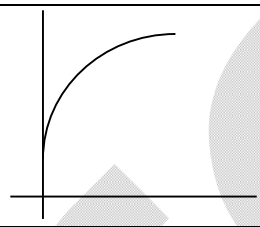
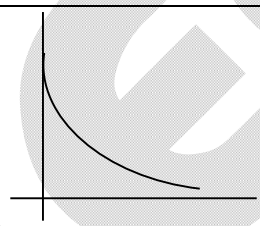
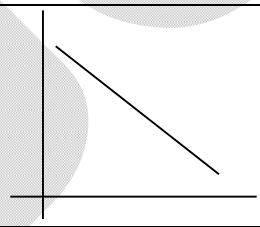
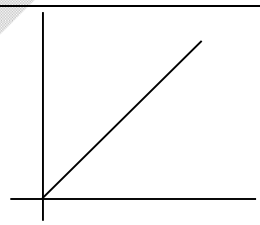
26. Match List – I with List – II:

List - I		List - II	
(P)	π_1 : 0.1 M glucose; π_2 : 0.1 M urea	(1)	π_1 and π_2 are isotonic
(Q)	π_1 : 0.1 M NaCl; π_2 : 0.1 M Na_2SO_4	(2)	No net migration of solvent across the semi permeable membrane
(R)	π_1 : 0.1 M NaCl; π_2 : 0.1 M KCl	(3)	π_1 is hypertonic to π_2
(S)	π_1 : 0.1 M $CuSO_4$; π_2 : 0.1 M sucrose	(4)	π_1 is hypotonic to π_2
		(5)	Both form azeotropic mixture

Codes:

	P	Q	R	S
(A)	1, 3	2	4	3
(B)	1,2	4	1,2	3
(C)	3	4	1, 2	5
(D)	1, 2	5	1, 2	4

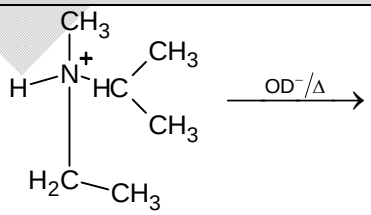
27. Match List – I with List – II:

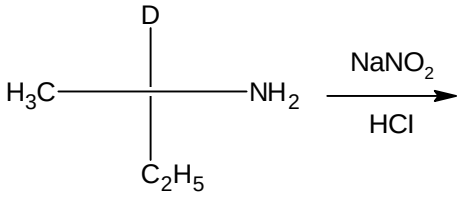
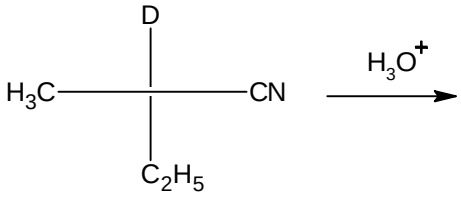
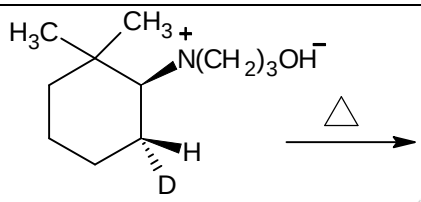
List - I (First order of reaction $A \rightarrow B$)		List - II	
(P)	[B] vs t	(1)	
(Q)	[A] vs t	(2)	
(R)	$\frac{1}{[A]}$ vs t	(3)	
(S)	In k vs $\frac{1}{T}$	(4)	
		(5)	

Codes:

	P	Q	R	S
(A)	1	3	2	4
(B)	2	5	3	4
(C)	4	3	1	2
(D)	1	4	5	2

28. Match the reaction given in List – I with the facts given in List – II:

List - I		List - II	
(P)		(1)	Develops a racemic mixture

(Q)	 (One enantiomer)	(2)	An alkene is obtained
(R)	 (One enantiomer)	(3)	Configuration is retained
(S)	 (One isomer)	(4)	Product may contain deuterium
		(5)	No stereogenic centre in the product

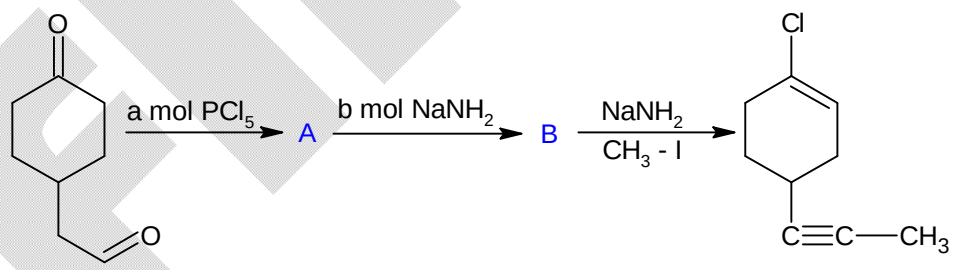
Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	1,2,4	1,4	3,4	2,5
(C)	1, 2	1, 4	3, 4	1, 5
(D)	2, 3	1, 4	1, 5	2, 5

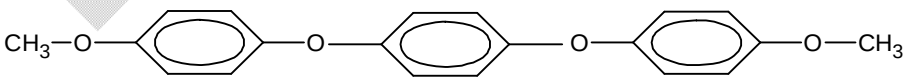
SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. 

Sum of a + b =

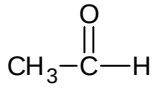
30. 

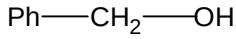
How many moles of HI consumed in above reaction?

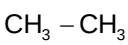
31. How many compounds give positive Tolle's Test ?

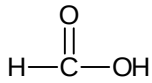
(A) D – Glucose

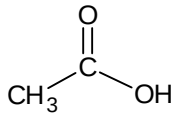
(B) D – Fructose

(C) 

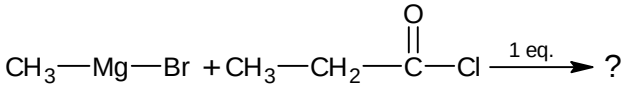
(D) 

(E) 

(F) 

(G) 

32. Total number of enol possible for the compound formed during given reaction will be (including stereoisomer) :



33. When β Amino butanoic Acid is heated how many diastereomers are obtained?

34. Total number of lone pairs of electrons in Melamine is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Let $A=\left[a_{ij}\right]$ be the set of matrices of order 3×3 with entries from the set {0, 1, 2, 3,4, 5, 6, 7, 8, 9} and $M=\left\{a_{ij}=0\forall i<j\right\}$ and N = {$|A| \neq 0$} are two events of the set A then which of the following probabilities hold(s) good-

(A) $P(M)=\frac{1}{10}$ (B) $P(M)=\frac{1}{1000}$

(C) $P\left(\frac{N}{M}\right)=\frac{9}{10}$ (D) $P\left(\frac{N}{M}\right)=\frac{729}{1000}$

36. The solution set of equation $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$ belongs to

(A) (−1, 0) (B) [−1,0]

(C) [−1, 1) (D) $\left[-\frac{1}{2},\frac{1}{2}\right]$

37. If $\log_x\left(y^{\log_3 x}\right)=\log_y\left(x^{\log_3 x}\right)$, where $x,y\in N$ and $\log_x\left(z-(x-y)^2\right)=4$, then:

(A) xy = 1 (B) x – y = 0

(C) Minimum value of (x + 3y) is 8 (D) Minimum value of z is 16

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. If n arithmetic means $a_1,a_2,a_3,.....,a_n$ are inserted between 50 and 200 and n harmonic means $h_1,h_2,h_3,.....,h_n$ are also inserted between the same two numbers, then the value of a_2h_{n-1} is equal to-

(A) 5000 (B) 10000

(C) $\frac{5000}{n}$ (D) $\frac{10000}{n}$

39. Fourteen digit numbers are formed by using the digits 1,2, 3, 4, 7, 7, 7, 7, 7, 7, 7, 7,7, 7 only where 1, 2, 3, 4 are called as distinct digits and 7, 7, 7,…., 7 (10 times) are called identical digits. If the number of such numbers where atleast two identical digits occur between any two distinct digits is k. 5! then the value of k is-

(A) 7 (B) 14

(C) 21 (D) 28

40. The value of $\int_a^b(x-a)^3(b-x)^4\,dx$ is $\frac{(b-a)^n}{m}$ then the value of (n, m)is

(A) (6,260) (B) (8,280)

(C) (4,240) (D) (7,300)

41. Four vectors $\vec{a},\vec{b},\vec{c}$ and $\vec{x}$ satisfy the relation $\left(\vec{a}\,.\,\vec{x}\right)\vec{b}=\vec{c}+\vec{x}$ where $\vec{a}\,.\,\vec{b}\neq 1$. The value of $\vec{x}$ in terms of $\vec{a},\vec{b}$ and $\vec{c}$ is equal to

(A) $\dfrac{\left(\vec{a}\,.\,\vec{c}\right)\vec{b}-\vec{c}\left(\vec{a}\,.\,\vec{b}-1\right)}{\left(\vec{a}\,.\,\vec{b}-1\right)}$ (B) $\dfrac{2\left(\vec{a}\,.\,\vec{c}\right)\vec{b}-\vec{c}}{\left(\vec{a}\,.\,\vec{b}-1\right)}$

(C) $\dfrac{\left(\vec{a}\,.\,\vec{c}\right)\vec{b}+\vec{c}\left(\vec{a}\,.\,\vec{b}-1\right)}{\left(\vec{a}\,.\,\vec{b}-1\right)}$ (D) $\dfrac{2\left(\vec{a}\,.\,\vec{c}\right)\vec{c}+\vec{b}}{\left(\vec{a}\,.\,\vec{b}-1\right)}$

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Match List – I with List – II:

List - I		List - II	
(P)	$f:R\to R, f(x)=\begin{cases}x^2+2mx-1 & x\le 0\\ mx-1 & x>0\end{cases}$ If $f(x)$ is one-one, then m can be	(1)	17
(Q)	If $f(x)=ax^7+bx^3+cx-5$; a, b, c are real constants and $f(-7)=7$, then value of $f(7)+17\sin x$ can be	(2)	34
(R)	If $f:R^+\to R^+$, is a polynomial function satisfying the functional equation $f(f(x))=6x-f(x)$, then $f(17)$ is equal to	(3)	1
(S)	If a polynomial function f satisfies the functional equation $f(x^2+1)=f^2(x)+1$ and $f(0)=0$, then $f(17)$ is equal to	(4)	-17
		(5)	51

Codes:

	P	Q	R	S
(A)	4	3	2	5
(B)	1	4	4	1
(C)	1	3	4	5
(D)	4	4	2	1

43. Match List – I with List – II:

List - I		List - II	
(P)	Maximum number of triangles formed by joining the point of intersection of the lines $y = x + 3$, $y = 2x + 3$, $y = 3x + 2$ and $x + y = 3$ is	(1)	1
(Q)	If the two intersecting lines intersect the hyperbola and neither of them is a tangent to it, then the maximum number of intersecting points is	(2)	2
(R)	If $ab^2c^3, a^2b^3c^4, a^3b^4c^5$ are in A.P. (a, b, $c > 0$), then the minimum value of ($a + b + c$) is	(3)	3

(S)	$25\int_0^{\frac{\pi}{4}}\left(\tan^6\left(x-[x]\right)+\tan^4\left(x-[x]\right)\right)dx$ is equal to (where [x] is greatest integer less than or equal to x)	(4)	4
		(5)	5

Codes:

	P	Q	R	S
(A)	4	5	1	1
(B)	3	4	3	5
(C)	4	5	1	1
(D)	3	4	5	3

44. If the focus and corresponding directrix of an ellipse having eccentricity $\frac{1}{2}$ are (3, 4) and $x + y = 1$, respectively, then

List - I		List - II	
(P)	End point of major axis is	(1)	(5, 7)
(Q)	If $2a$ is length of major axis and $2b$ is the length of minor axis of the ellipse then $\left(b^2,a^2\right)$ is	(2)	(4, 5)
(R)	Centre of the ellipse is	(3)	(6, 7)
(S)	Other focus is	(4)	(5, 6)
		(5)	(6, 8)

Codes:

	P	Q	R	S
(A)	4	2	3	4
(B)	3	5	1	2
(C)	4	2	1	2
(D)	3	5	2	4

45. Match List – I with List – II:

List - I		List - II	
(P)	If f: [1, 3] → [1, 3] is a continuous decreasing function, then $\int_1^3\left(f(x)-f^{-1}(x)\right)dx$ is equal to	(1)	0
(Q)	The maximum value of $\left(\frac{5-\cos 3x}{3+\cos 5x}\right)$ is	(2)	2
(R)	Let $f:R\to R$ be a function defined by $f(x)=x^3+px^2+qx-3$. If f is monotonically decreasing in the interval (1, 3) only, then (p + q) is equal to	(3)	3
(S)	If $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{n}{(n+r)(2n+r)}=\ln\left(\frac{a}{b}\right)$ where a and b are co-prime, then $\lvert a+b\rvert$ is equal to	(4)	4
		(5)	7

Codes:

	P	Q	R	S
(A)	1	3	3	2
(B)	4	4	2	5
(C)	1	4	3	5
(D)	4	3	2	2

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Let z_1 and z_2 be distinct randomly chosen roots of the equation $z^{100} = 1$. If the probability that $|z_1 + z_2| \geq \sqrt{2+\sqrt{3}}$ is expressed as $\frac{p}{q}$ (where p and q are in their lowest form), then value of $(p + q)$ is

47. Let $f(x) = x + \sqrt{1-x^2}$. Find the number of symmetric matrices A of order equals to the number of integers in the range of $f(x)$ and whose elements are the integral elements in the range of $f(x)$ and also $tr(A)$ is either equal to 0 or 1.

48. If two circles $C_1 : x^2 + (y-4)^2 = 12$ and $C_2 : (x-3)^2 + y^2 = 13$ intersect at points A and B and a quadrilateral ACBD is formed by tangents at A and B to both circles then, the diameter of circumcircle of quadrilateral is

49. If $x = 4t^3 + 3, y = 3t^4 + 4$ and $\frac{\left(\frac{d^2x}{dy^2}\right)}{\left(\frac{dx}{dy}\right)^n}$ is a constant, then find the value of series

$1 + \frac{4}{5} + \frac{4}{5n} + \frac{4}{5n^2} + \ldots\ldots^{\infty}$

50. If the area of the region $\left\{(x,y) \in R^2 : y^2 \geq 4x, |y| \leq \frac{x}{2} + 2\right\}$ is Δ, then the value of [Δ] is, (where $[x]$ is greatest integer less than or equal to x):

51. $a^2 \cot 9^0 + b^2 \cot 27^0 + c^2 \cot 63^0 + d^2 \cot 81^0$. Let a, b, c and d be real numbers satisfying $a + b + c + d = 5$. If the minimum value of the expression above is equal to $\frac{\sqrt{x}}{y}$, where x and y are co-prime positive integers, then then the value of $(x + y)$.